

TIC TAC TOE GAME

Project Report

Name: JANVEE KUMARI

Reg No: 26MIP10060

Course: DATA SCIENCE

University: VIT Bhopal University

Python Project • First Year CSE

1. Introduction

This project is about understanding and building game logic using Python. Tic Tac Toe is used as a simple game to learn how a program can follow rules, manage turns, take user input, and decide the result.

The project helps in understanding basic programming concepts such as lists, functions, loops, conditions, and input/output.

2. Problem Statement

The main problem is to understand how game logic works and build it using simple Python code. Tic Tac Toe is used to apply this logic. The program must manage player turns, accept moves, check whether a move is valid, identify winning combinations, and detect a draw.

3. Functional Requirements

- Show a 3 × 3 game board.
- Allow Player X and Player O to enter moves.
- Accept positions from 1 to 9.
- Check if a position is already used.
- Check for a winning combination.
- Check if the game is a draw.
- Display the final result.

4. Non-functional Requirements

- The program should be easy to use.
- The program should give clear messages for wrong input.
- The program should run in a normal Python environment.
- The code should be simple and easy to understand.
- The program should respond quickly to user input.

5. System Architecture

The project is a small Python console application. It does not use a database or web server. The main program contains the board and functions for displaying the board, checking a winner, and checking a draw.

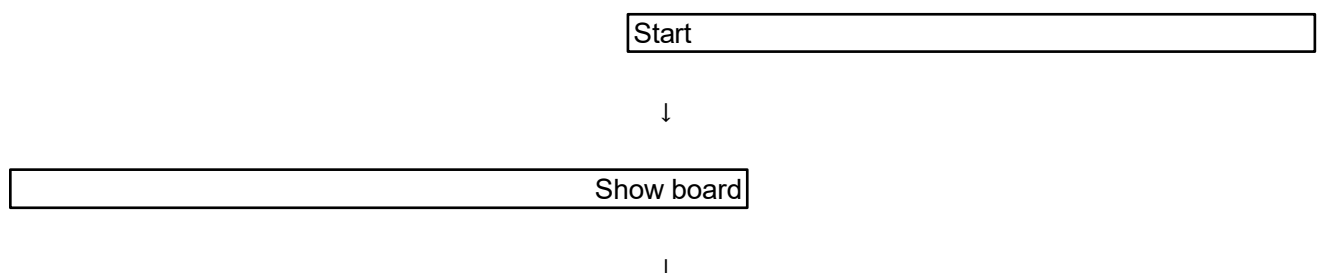
Player	Python Game Program	Terminal Output
Input position 1–9	Board + Game Logic	Board + Result

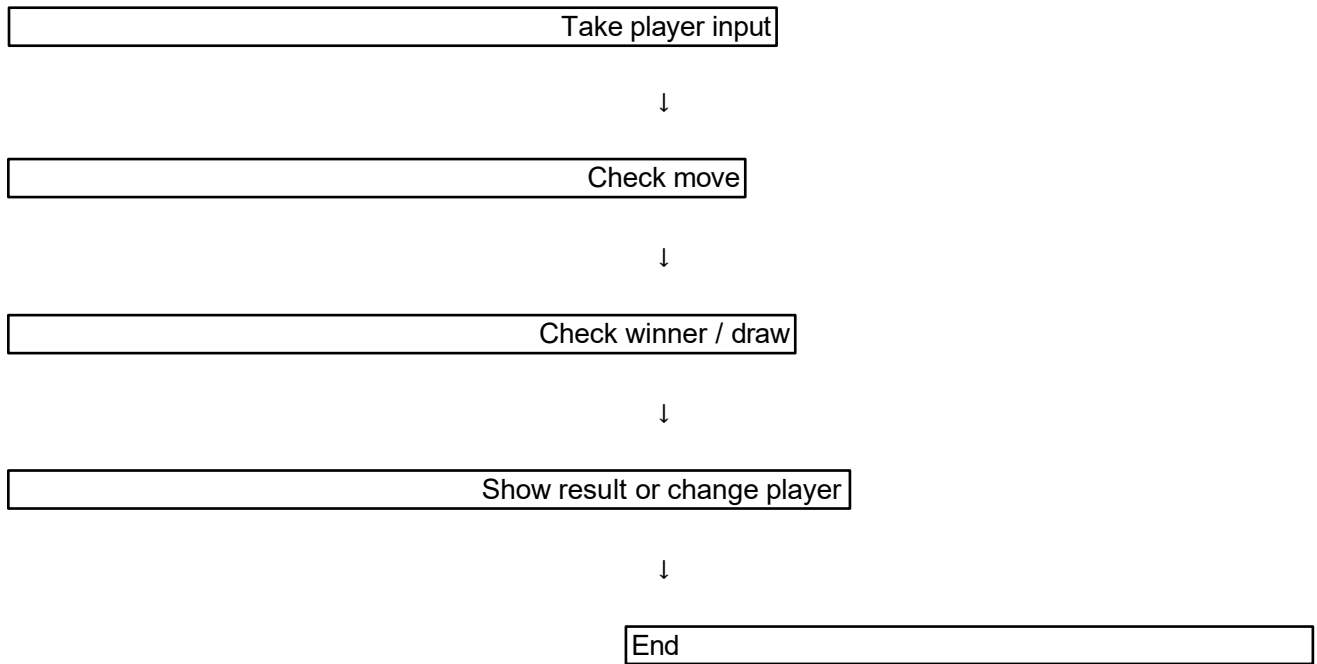
6. Design Diagrams

6.1 Use Case Diagram

Player 1	Play Game	Player 2
Choose position	View board Check result	Choose position

6.2 Workflow Diagram





6.3 Sequence Diagram

Player	Game	Board
Enter position	Receive input	Update position
	Check winner	Return board state
View result	Show result	

6.4 Class / Component Diagram

Tic Tac Toe Program
board list
show_board()
check_winner(player)
check_draw()
main game loop

6.5 ER Diagram

An ER diagram is not needed because the game does not store data in a database. The board exists only while the program is running.

7. Design Decisions & Rationale

- A list of 9 spaces is used for the board because it is simple.
- Functions are used to keep the code organised.
- A list of winning positions is used to check all possible wins.

- The terminal is used because the project is meant to be simple and easy to run.
- No database is used because the game does not need permanent data.

8. Implementation Details

The program starts with an empty board. The `show_board()` function prints the board. The `check_winner()` function checks the eight possible winning combinations. The `check_draw()` function checks whether there are no empty spaces left. The main while loop controls the turns of X and O.

9. Screenshots / Results

The program displays a 3 × 3 board in the terminal. After a winning move, it prints a message such as "Player X wins!". If all spaces are filled without a winner, it prints "It's a draw!".

10. Testing Approach

Test	Expected Result
Enter a number from 1 to 9	Move is placed
Enter a used position	Program shows an error message
Enter a number outside 1 to 9	Program asks for a valid position
Enter letters	Program asks for a number
Make three symbols in a row	Player wins
Fill board without a winner	Game is a draw

11. Challenges Faced

- Understanding all possible winning combinations.
- Handling wrong user input.
- Making sure a used position cannot be selected again.
- Changing turns correctly between X and O.

12. Learnings & Key Takeaways

- Learned how to use lists in Python.
- Learned how functions make code easier to manage.
- Practiced if-else conditions and loops.
- Learned how to take input from users.
- Learned how to build and test simple game logic.

13. Future Enhancements

- Add a single-player mode against the computer.
- Add a score counter.
- Add a graphical user interface.
- Add a restart option.

- Save game scores for future games.

14. References

- Python 3 documentation and basic Python learning material.
- Class notes and lab practice.
- VIT Bhopal project submission requirements.
- Vityarthi course